

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester IV)
Dot .Net Core and Progressive Web Application Practical

Practical – VII

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Class: TYIT	Batch: 1
Date of Assignment: 04/08/2026	Date/Time of Submission: 04/08/2026

Aim(A) Develop a PWA that works in offline mode using Service Worker.

CODE :

index.cshtml :

```
@{
    ViewData["Title"] = "Collage Portal";
}
<div class="text-center">
    <h1 class="display-4">Collage Website</h1>
    <h2>Time Table Download</h2>
    <h2>Notices Download</h2>
    <h2>Study Material Download</h2>
</div>
```

Layout.cshtml :

```
<link rel="manifest" href="/manifest.json" />
```

[Before Head closing add this part]

```
<script>
    if ('serviceWorker' in navigator) {
        navigator.serviceWorker.register('/service-worker.js');
    }
</script>
```

[Before Body closing add this part]

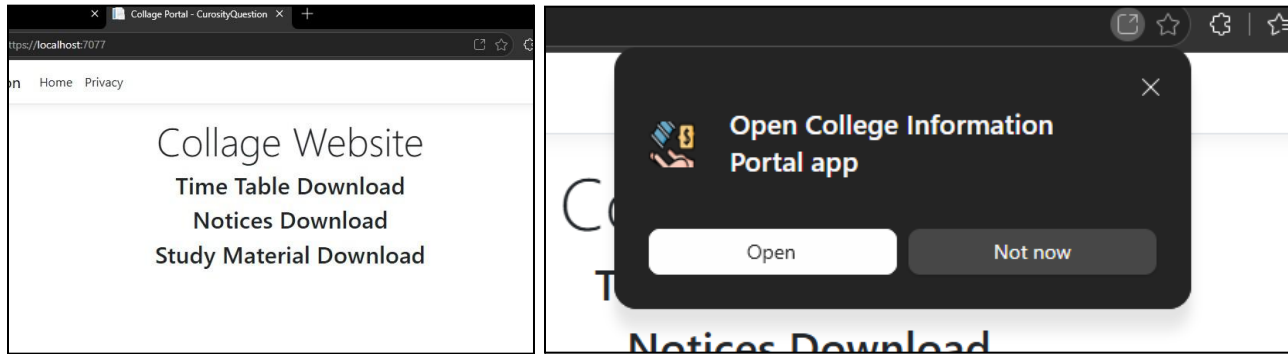
service-worker.js

```
self.addEventListener("install", event => {
    console.log("Service Worker Installed");
});
self.addEventListener("fetch", event => {
    event.respondWith(
        fetch(event.request)
    );
});
```

manifest.json :

```
{
    "name": "College Information Portal",
    "short_name": "College Portal",
    "start_url": "/",
    "display": "standalone",
    "icons": [
        {
            "src": "icons/icon1.png",
            "sizes": "128x128",
            "type": "image/png"
        },
        {
            "src": "icons/icon2.png",
            "sizes": "512x512",
            "type": "image/png"
        }
    ]
}
```

OUTPUT :



Offline Page Code Changes :

service-worker.js :

```
const CACHE = "v1";

self.addEventListener("install", e => {
  e.waitUntil(
    caches.open(CACHE).then(cache => {
      return cache.addAll([
        "/",
        "/offlinePage.html"
      ]);
    })
  );
});

console.log("Service Worker Installed");
});

self.addEventListener("fetch", e => {
  e.respondWith(
    fetch(e.request).catch(() =>
      caches.match("/offlinePage.html")
    )
  );
});
```

OfflinePage.html :

```
<!DOCTYPE html>
<html>
<head>
  <meta charset="utf-8">
  <title>Offline</title>
</head>
<body>

  <h1>Welcome to offline College Portal </h1>

  <p>Student Details </p>
  <p>Name :Som Mishra </p>
  <p>Roll.No. : T023</p>
  <p>Department : Information Techonlogy </p>
  <hr />
  <h1>College Dashboard</h1>
  <p>Time Table Download</p>
  <p>Notices Download</p>
  <p>Study Material Download</p>
  <hr />
  <p>Please check your internet connection and try again.</p>

</body>
</html>
```

OUTPUT :

Welcome to offline College Portal

Student Details

Name : **Som Mishra**

Roll.No. : **T023**

Department : Information Techonlogy

College Dashboard

Time Table Download

Notices Download

Study Material Download

Please check your internet connection and try again.

Aim(B) : Create a Student Record Management PWA. Additional Code :

Student.cs :

```
namespace CurocityQuestion.Models
{
    public class Student
    {
        public int RollNo { get; set; }
        public string Name { get; set; }      omkar sawant
    }
}
```

Index.cshtml :

```
@{
    ViewData["Title"] = "Collage Portal";
}
<div class="text-center">
    <h1 class="display-4">Collage Website</h1>
    <h1>Offline Page</h1>
    <h2>Student Record</h2>
    <p>Roll No : @Model.RollNo</p>
    <p>Name : @Model.Name</p>
</div>
```

service-worker :

```
const CACHE = "v1";

self.addEventListener("install", e => {
    e.waitUntil(
        caches.open(CACHE).then(cache => {
            return cache.addAll([
                "/",
                "/Home/Index",
                "/css/site.css",
                "/js/site.js"
            ]);
        })
    );
});
```

```
.  
addEventListener("fetch", e => {  
    e.respondWith(  

```

```
    caches.match(e.request).then(response => {  
        return response || fetch(e.request);  
    })  
    });  
});
```

HomeController.cs :

```
Student s = new Student();  
s.RollNo = 40;  
s.Name = "Omkar";
```

[add after public IActionResult Index()]

```
return View(s);
```

OUTPUT :

Collage Website
Offline Page
Student Record

Collage Website
Offline Page
Student Record